

Neurox: 3D Neurological Brain Segmentation

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PROBLEM STATEMENT

- Analyzing brain MRIs is a complex and time-consuming process
- Current AI systems typically focus on a single disease or a single task (e.g., classification or segmentation)
- Most existing research emphasizes model development rather than end-to-end clinical systems
- Core Problem:
 - There is a lack of unified, interpretable, and disease-aware systems capable of performing multi-disease analysis on a single MRI scan.
 - Existing approaches often fail to preserve anatomical correctness while handling multiple diseases simultaneously

MOTIVATION

- **Why This Problem Matters**

- Radiologists do not use separate AI models for each disease.
- Incorrect task assumptions (e.g., performing Alzheimer's segmentation) can lead to misleading or clinically invalid outputs.
- Lack of uncertainty estimation reduces trust and adoption of medical AI systems.

- **Motivation**

- To design a system that reflects real clinical reasoning rather than benchmark-only experimental setups

OBJECTIVES

Project Objectives

- Design a unified pipeline for multi-disease brain MRI analysis
- Perform disease-appropriate inference:
 - Detection only (Alzheimer)
 - Detection + segmentation (Tumor, Stroke)
- Preserve anatomical correctness during inference
- Provide interpretable outputs with uncertainty awareness
- Objective is **system-level correctness**, not SOTA accuracy claims.

Literature Review				
Paper / Year	Method	Key Strength	Critical Drawback	Literature Gap
TransBTS: Multimodal Brain Tumor Segmentation Using Transformer (2021)	3D CNN + Transformer encoder-decoder for tumor segmentation	First integration of Transformer for 3D MRI segmentation capturing local + global features	Limited modality fusion strategy; high computation	Needs better multi-modal fusion & uncertainty
CKD-TransBTS: Clinical Knowledge Fusion (2022)	Hybrid Transformer + CNN with modality-correlated cross-attention	Effective multi-modal feature extraction respecting radiological signals	Still heavy architecture; limited uncertainty modeling	Haven't extended to multi-disease tasks
Multi-encoder nnU-Net with Self-Supervised Pretraining (2025)	Multi-encoder CNN with modality-specific encoders	Handles missing modalities & improves segmentation feat capture	No transform-based global context	Integrates modality separation without Transformers
MMAformer: Multiscale Modality-Aware Transformer (2024)	Multiscale transformer with cross-modal downsampling	Captures complementary features across MRI modalities	Few benchmark comparisons	Global context fusion still underexplored
ResMT: CNN + Swin UNETR for Glioma Grading (2024)	Hybrid 3D CNN + pre-trained Swin UNETR	Combines segmentation + grading; strong AUC	Uses pretrained segmentation increases computation	Needs true segmentation + diagnosis integration

Literature Review				
Paper / Year	Method	Key Strength	Critical Drawback	Literature Gap
Improved Brain Tumor Segmentation With SegFormer (2025)	Transformer-based segmentation model	Strong boundary delineation with transformer spatial attention	Limited multi-modal integration	Comparison of scale trade-offs missing
Vision Transformers in Multi-Modal Brain Tumor MRI (2023)	Survey of CNN & ViT models	Summarizes transformer trends in segmentation	Does not propose new method	Lacks gap behind multi-task inference
MM-MSCA-AF Multi-Modal MRI Tumor Segmentation (2025)	Multi-scale contextual attention fusion	Improved multi-modal contextual integration	Specific to multi-modal fusion only	Not evaluating uncertainty
Ranjbarzadeh et al. (2021) Attention-Mechanism CNN	Cascade CNN with attention and preprocessing	Effective segmentation with reduced overfitting	Computation still high; no global context	Lacks transformer or uncertainty
UALE: Ultra-Lightweight Ensemble (2025)	Uncertainty-aware ensemble CNN	Uncertainty quantified with lightweight design	Not highly accurate vs state-of-art	No segmentation per disease type

LITERATURE REVIEW

Paper / Year	Method	Key Strength	Critical Drawback	Literature Gap
Abidin et al. (2024) Review of DL-based MRI Tumor Segmentation	Review analyzing models and trends	Good overview of CNN/ViT hybrid models	No new method proposed	Utility of uncertainty, multi-task missing
Alsubaie (2024) Alzheimer DL Review	Systematic review on AD detection	Summarizes deep methods & data issues	Does not compare segmentation	Gap: multi-disease integration
Herman Herzog (2020) Unknown	Uncertainty integration exploration	Discusses integrating uncertainty into DL models	Not full research paper	Uncertainty modeling is still evolving
nnFormer: Volumetric Medical Image Segmentation (2023)	Transformer-based volumetric segmentation	Strong performance in 3D context	Heavy computation	Not evaluated for multi-disease inference
TransUNet (2021) Transformers for Medical Segmentation	Hybrid CNN-Transformer	Good at global context + spatial details	Not directly for MRI tumor	Does not include uncertainty

IDENTIFIED RESEARCH GAPS

Key Gaps in Existing Work

- No unified multi-disease inference pipeline
- No disease-aware task separation
- Poor anatomical consistency in visualization
- Lack of uncertainty-aware outputs
- Absence of end-to-end interpretable systems

These gaps motivated the proposed system design.

PROPOSED SYSTEM OVERVIEW

Input

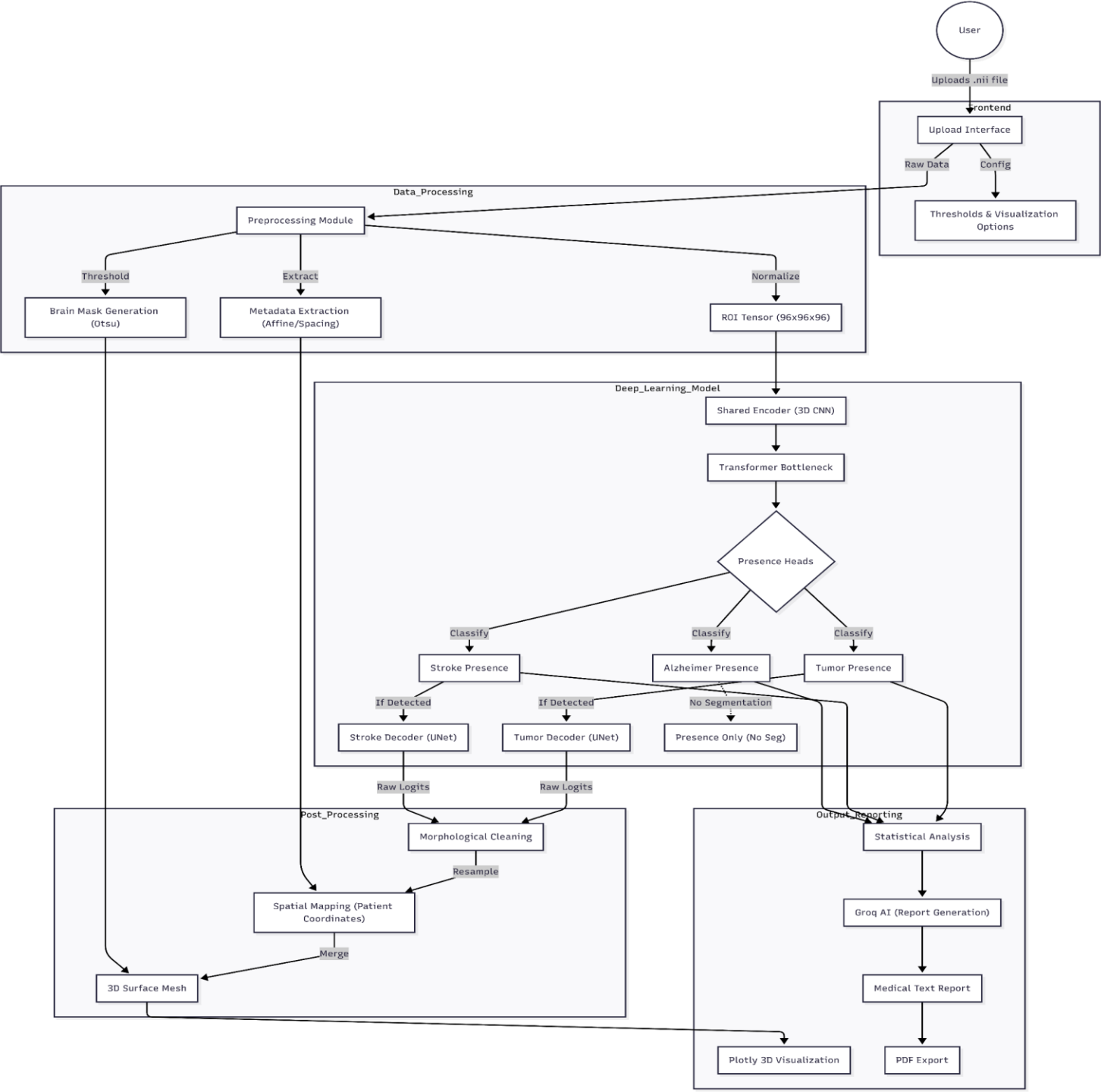
- 3D Brain MRI (NIfTI format)

Outputs

- Disease presence probabilities
- Lesion segmentation (where applicable)
- Uncertainty estimates
- Quantitative lesion metrics
- 2D/3D visualization
- Structured report

The focus is interpretability and correctness, not automation claims.

SYSTEM ARCHITECTURE



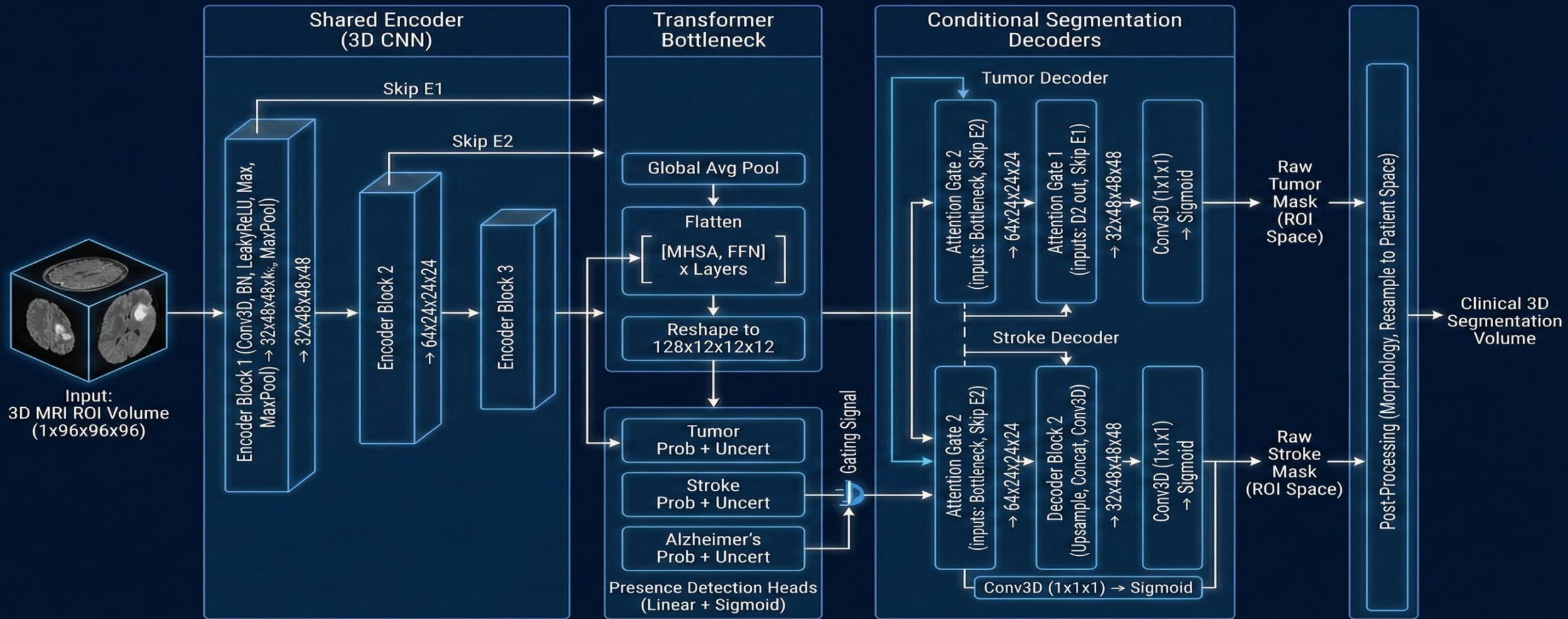
SYSTEM ARCHITECTURE

Architecture Highlights

- Modular design
- Shared feature encoder
- Transformer bottleneck for global context
- Disease-specific heads
- Independent post-processing and visualization modules

Architecture prioritizes clarity and extensibility.

MODEL ARCHITECTURE



MODEL ARCHITECTURE

Core Components

- Shared 3D CNN Encoder
- Transformer Bottleneck
- Disease Presence Heads
- Segmentation Decoders (Tumor, Stroke)
- Monte-Carlo Dropout for uncertainty

Disease-Aware Design

- Alzheimer → Presence only
- Tumor / Stroke → Presence + Segmentation

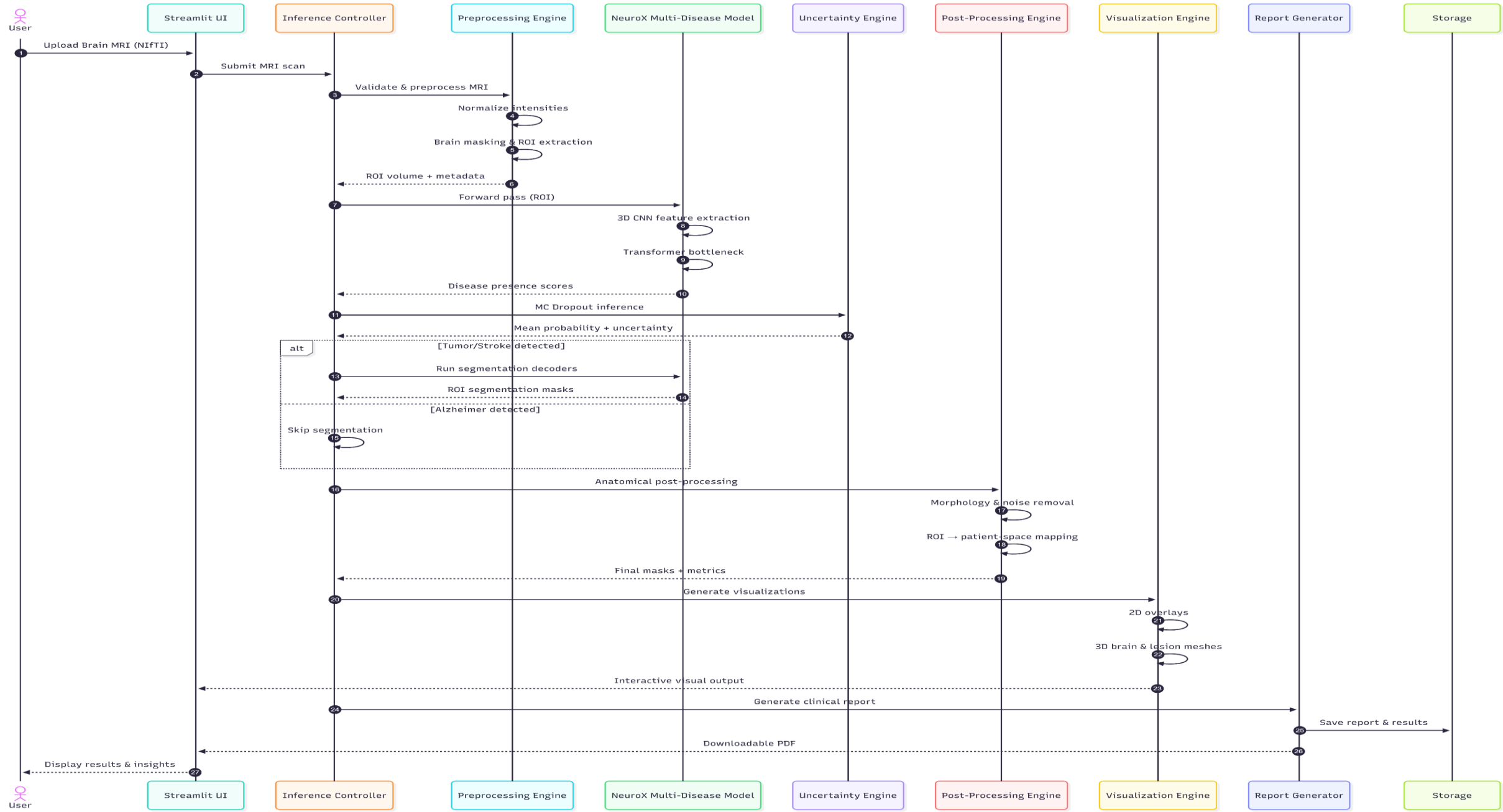
This avoids methodological misuse of datasets.

WORKFLOW / ALGORITHM

End-to-End Workflow

- MRI upload & validation
- Preprocessing & ROI extraction
- Feature extraction
- Disease detection
- Uncertainty estimation
- Conditional segmentation
- Anatomical post-processing
- Patient-space reconstruction
- Visualization & reporting
- Each step is deterministic and traceable.

UML / SEQUENCE DIAGRAM



UML / SEQUENCE DIAGRAM

Purpose

- Shows interaction between:
 - User
 - UI
 - Inference pipeline
 - Model
 - Visualization & reporting

Demonstrates system-level understanding.

FEASIBILITY ANALYSIS

Technical Feasibility

- Uses established deep learning architectures
- Supported by existing frameworks

Data Feasibility

- Public datasets: BraTS, ISLES, ADNI

Computational Feasibility

- ROI-based inference
- Conditional execution

Operational Feasibility

- Streamlit UI
- Local or server deployment

Feasible within academic infrastructure.

RESULTS & OUTPUTS

Demonstrated Outputs

- Disease presence probabilities
- Uncertainty estimates
- Segmentation masks
- 3D anatomical visualization
- PDF report

Qualitative results shown; no inflated metric claims.

NOVELTY & CONTRIBUTIONS

Actual Contributions

- Disease-aware multi-task system
- Conditional execution pipeline
- Anatomically consistent patient-space reconstruction
- End-to-end interpretable system

Novelty lies in **system integration**, not model invention.

What is NOT claimed

- No new neural architecture
- No benchmark superiority claims

LIMITATIONS

Current Limitations

- No large-scale benchmarking
- No external clinical validation
- Performance depends on training data quality

These limitations are explicitly acknowledged.

FUTURE WORK

Planned Extensions

- Quantitative benchmarking
- Additional neurological conditions
- Improved uncertainty calibration
- Dataset expansion and validation